

PAPER_1743 — Beal Conjecture = $A^{x+B}y=C^z$ ($x,y,z \geq 3$) requires $\gcd(A,B,C) > 1$ (\$1M Beal prize)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Number Theory **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_1271-1299 broader corpus)

Observation

PAPER_1296 (PAPER_1271-1299 broader corpus) gives:

$A^x+B^y=C^z$ ($x,y,z \geq 3$) requires $\gcd(A,B,C) > 1$ (\$1M Beal prize)

Status: **EXACT**.

UQFF Closed Identity

$A^x+B^y=C^z$ ($x,y,z \geq 3$) requires $\gcd(A,B,C) > 1$ (\$1M Beal prize) EXACT

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1296
- Related: PAPER_1716-1725 (tier-30 Millennium); PAPER_1726-1735 (tier-31 neutron lifetime)
- Calculator dispatch: `calculate_paradox({"paradox": "beal_gcd_gt_1"})`

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